Focus of this homework:

- Fundamentals of free vibration of damped SDOF vibratory systems
- Using computer (e.g., MATLAB) to assist in engineering design problems

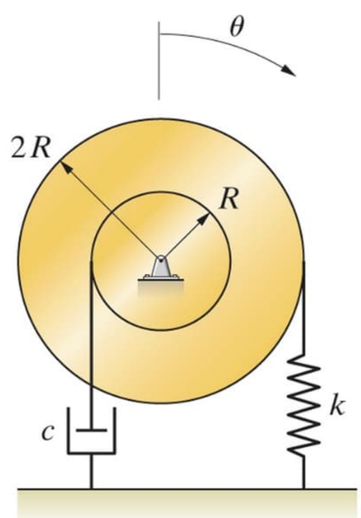
Submit homework (in hard copies) by the beginning of the class. It is your responsibility to make sure that homework is submitted in good quality. Please organize your work neatly. Suggest to start each problem on a new page.

0) As a start to understand damping in structures: (a) list some examples of physical devices and systems that can be approximately modeled as viscous dampers (damping force is linearly proportional to the velocity); (b) Using open sources (such as baidu) or similar search engines, textbooks, etc., find the damping ratios of structural materials (metals, etc.).

This problem is intended for self-study. Most of the solution can be found in the notes.

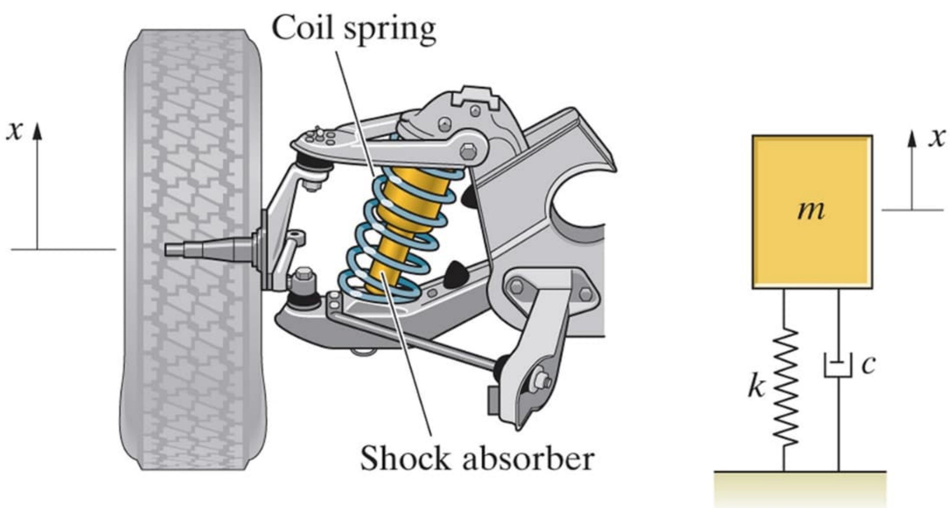
1) Consider the following system with $R = 250$ mm, $k = 150$ N/m, and the moment of inertia of the disk is $I = 2$ kg-m^2.

   a) Derive the governing equation of motion using the energy method and the dissipation function.
   b) At what value of c is the system critically damped?
   c) At $t = 0$, the spring is unstretched and the clockwise angular velocity of the disk is 10 rad/s. If the system is critically damped, determine the response $\theta(t)$ as a function of time. Plot this response graphically.
   d) Determine the maximum resulting angular displacement of the disk and the time at which it occurs. You can do this numerically if you prefer.
   e) Determine (numerically or graphically) the time of the first occurrence for the system to reduce its response to 10% of the maximum amplitude.
   f) Change one physical parameter and repeat part (e) to understand the concept of *parametric design and optimization* commonly employed in the industry.

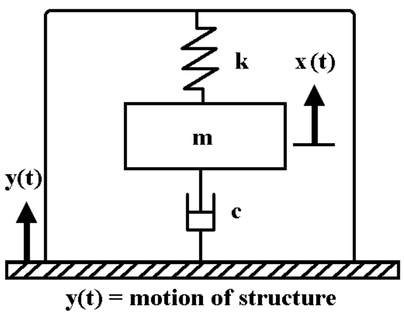


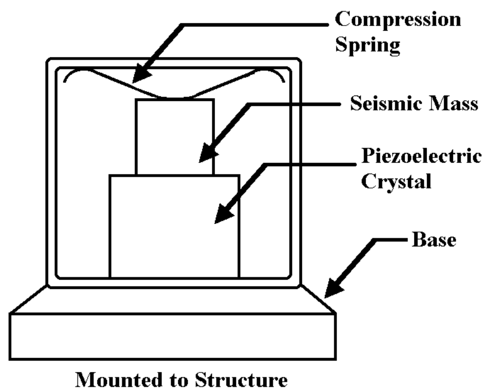
2) The motion of the car's suspension can be modeled by the damped spring-mass oscillator with $m = 36$ kg, $k = 22$ kN/m, and $c = 2.2$ kN-s/m. Assume that no external forces act on the tire and wheel. At $t = 0$, the spring is unstretched and the tire and wheel are given a vertical velocity $dx/dt = 10$ m/s.

   a) Determine the response $x(t)$ as a function of time. Plot this response graphically.
   b) Determine (numerically) the time it takes for the system to reduce its amplitude to 10% of the maximum amplitude.

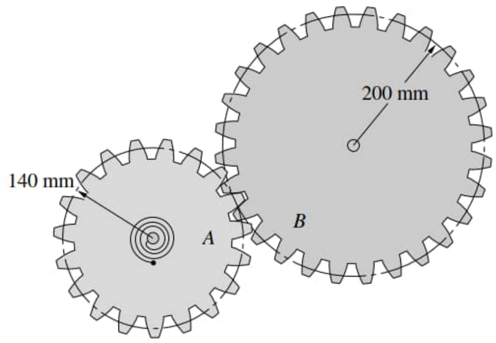


3) An accelerometer has a suspended mass of 0.01 kg with a damped natural frequency of vibration of 100 Hz. When mounted on an engine undergoing an acceleration of 1 g at an operating speed of 6000 rpm, the acceleration is recorded as 10 $\mathrm{m/s^2}$ by the instrument. Find the damping constant and the spring stiffness of the accelerometer (Choose a damping ratio close to 0.6 if possible).

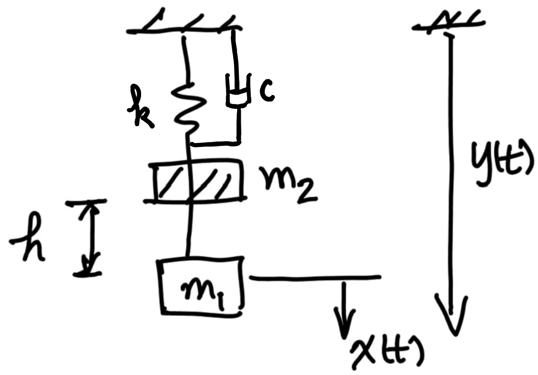




4) The moments of inertia of *A* and *B* are $I_A = 0.025$ kg-m^2 and $I_B = 0.1$ kg-m^2. Gear *A* is connected to a torsional spring with constant $k = 10$ N-m/rad. The bearing supporting gear *B* incorporates a damping element that exerts a resisting moment on gear B of magnitude $2d\theta_B/dt$ N-m, where $d\theta_B/dt$ is the angular velocity of gear *B* in rad/s. What is the frequency of small angular vibrations of the gears?



5) A mass m_1 is in static equilibrium position when a mass m_2 is dropped a height of h and impacts m_1. After impact, there is no rebound (i.e., the coefficient of restitution $e = 0$). The system is supported by a spring and a damper. Solve for the response of the system.



6) The measured free vibration response of a one-degree-of-freedom system is shown.
   a) Deduce from the graph the log decrement, the natural frequency, and the critical damping ratio.
   b) Estimate the value of t beyond which the displacement magnitude will not exceed 0.01 mm.
   c) If the damping constant is held fixed, while the stiffness is doubled and halving the mass, how would that alter the answers to part (b)?
   d) If the initial displacement is $q_0 = -10$ mm, what is the initial velocity?